

SWE 380 / CSC 580 Assignment 1: GitHub + AI Engineering Practice

Assignment Overview

In this assignment, you will practice a professional workflow for using **GitHub** and **generative AI** together. You will start from an existing public portfolio website template, create your own repository, use an AI assistant to plan and implement a small set of changes, review the resulting code, publish the site with GitHub Pages, and document what you learned.

The objective is not to prove that AI can produce a website. The objective is to demonstrate that **you can direct, inspect, test, revise, and take responsibility for AI-assisted software work**.

Part 1. Prepare a Project Context and Work Repo

1. Choose your content source

You may use one of the following options:

- Your public professional resume.
- Your public professional profile (like on LinkedIn, CareerBuilder, or Indeed, etc.)
- A public professional profile, provided that you have permission to reuse the information.

PS: It is highly recommended to use UM-provided Gen-AI tools, as they are FERPA-compliant¹. Yet, if you do not want to use your public information, you can provide fictitious information that represents your skills.

Create a local text file named content-notes.md containing the information you intend to use. Include only information that you are comfortable publishing on a public website.

¹ <https://studentprivacy.ed.gov/ferpa>

2. Fork or copy the starter project (8 points)

First, obtain the repository in which you will complete and submit this assignment. Use the instructor-approved starter repository. If no alternative is provided, you may use the portfolio template referenced in the source activity: [vCard Personal Portfolio](#).

Fork the starter repository into your own GitHub account. You will use this repo as your template to update and customize. Configure the repository so that:

- The repository has a meaningful name.
- The repository includes a useful README.md.
- The repository is public only if you are comfortable publishing the work.
- Your content-notes.md does not contain private information.
- The original project license and attribution are preserved where required.

3. Write initial acceptance criteria (5 points)

Before asking AI to change the project, write at least five acceptance criteria. Each criterion must describe observable behavior.

Example:

- The landing page displays my name and a concise professional role.
- The portfolio contains at least two projects with a title and description.
- The navigation links move to the correct sections.
- The page is readable on a narrow mobile viewport.
- No placeholder name, email address, or image remains in the published site.

#	Acceptance criterion	How I will verify it
1		
2		
3		
4		
5		

4. Create a project issue (5 points)

Create a GitHub issue in your repository titled:

Build and publish my AI-assisted portfolio update

The issue must contain:

- A short problem statement.
 - The intended audience for the site.
 - The acceptance criteria from Question 2.
 - Any constraints, such as “This assignment will not add a database or login system.”
 - A checklist of major tasks.
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Part 2. Create Your GitHub Workflow

5. Create a feature branch — 7 points

Create a branch named:

```
feature/portfolio-personalization
```

Do not make the main-branch changes directly. On the feature branch, make one small commit that updates only the project content or configuration needed to begin the assignment.

Your commit message should be specific, for example:

```
Add initial portfolio content and project issue
```

Record the branch name and commit URL in your submission.

Part 3. Use AI as a Coding Assistant

6. Write a context-specific AI prompt (10 points)

Use an UM AI assistant (e.g., UMGPT, Gemini, Jules) of your choice. You may use the assistant to analyze the repository, propose a plan, generate content, explain code, or suggest tests.

Your prompt should include:

- The project purpose (building an online portfolio website).
- The files or sections that may be changed.
- The acceptance criteria.
- Technical constraints, such as preserving the existing layout and avoiding a backend.
- A request for a plan before implementation.
- A request to identify assumptions and risks.

You may adapt the following prompt, or use your own:

You are assisting with a small, public static portfolio website.

Project context:

- Purpose: present a student's skills and projects to internship reviewers.
- Existing system: an HTML/CSS/JavaScript portfolio template.
- Allowed changes: content, styling, and small front-end corrections only.
- Do not add: a backend, login system, analytics tracker, or external data collection.

Acceptance criteria:

1. [paste your criteria]
2. [paste your criteria]
3. [paste your criteria]
4. [paste your criteria]
5. [paste your criteria]

Constraints:

- Do not invent education, work history, awards, links, or technical skills.
- Preserve required license and attribution.
- Keep the site usable on mobile screens.
- Maintain readable contrast, keyboard navigation, meaningful link text, and useful image alternative text.
- Do not expose private information.

First provide:

1. A file-by-file implementation plan.
2. Any Assumptions or risks.

3. A proposed test checklist.
4. Any questions that must be answered before implementation.

Do not modify files until the plan is reviewed.

Save the prompt and the AI response in [ai-log.md](#). If the response is long, summarize it and include a link or dated excerpt where permitted.

7. Review the AI plan (5 points)

Before applying changes, identify:

- Parts of the plan you accepted.
- Any part you revised.
- Any risk or assumption the AI identified.
- Any risk or assumption it failed to identify.

Review item	Your response
Accepted plan element	
Revised plan element	
Risk the AI identified	
Risk the AI missed	

8. Implement in small changes (10 points)

Use the AI assistant to implement the approved changes. Work incrementally rather than requesting an uncontrolled rewrite of the entire repository.

Create at least **three meaningful commits** on the feature branch. Suggested sequence:

Update portfolio identity and navigation

Add projects and experience content

Improve responsive layout and accessibility

Each commit should be understandable independently. Do not commit passwords, API keys, tokens, private documents, or unnecessary generated files.

Record the commit history URL (part of your deliverables).

Part 4. Inspect, Test, and Review

9. Perform a human code review (10 points)

Inspect every AI-generated change before merging. Choose at least three files or code regions and record what you checked.

File region or	What the AI changed	What I verified	Change made after review if applicable
1.			
2.			
3.			

Your review must address correctness, unnecessary changes, invented content, or any other concerns.

10. Test the site (10 points)

Run the site locally or through the repository preview. Test at least the following:

Test	Result and evidence
Page loads without a visible error	
Navigation links reach the intended sections	
Personal content replaces template placeholders	

Test	Result and evidence
Project links and external links work	
Layout remains usable on a narrow viewport	
Keyboard navigation is usable	
Images have meaningful alternative text or are decorative	
No secrets or private data are exposed	
Browser console has no unexplained errors	
The implementation satisfies the five acceptance criteria	

Include screenshots or a short screen recording for at least three tests (part of your deliverables).

11. Open a pull request (8 points)

Open a pull request from [feature/portfolio-personalization](#) into the default branch.

The pull request description must include:

- Summary of the change.
- Link to the original issue.
- Acceptance-criteria checklist.
- Testing performed.
- AI assistance used.
- Known limitations or follow-up work.

Record the pull-request URL (part of your deliverables).

12. Merge into the main branch (5 points)

Merge the pull request into the default branch. Do not delete the feature branch.

Record the merge commit or repository history URL (part of your deliverables).

Part 5. Publish with GitHub Pages

13. Enable GitHub Pages (7 points)

Step 13.1: Enable GitHub Pages

1. Go to your repository on GitHub (e.g., `github.com/your-username/my-portfolio`)
2. Click on the **Settings** tab (gear icon)
3. In the left sidebar, scroll down and click on **Pages**
4. Under "Build and deployment," find the **Source** section
5. Select **Deploy from a branch**
6. Under **Branch**, select `main` (or `master`) and `/` (`root`)
7. Click **Save**

Step 13.2: Wait for Deployment

GitHub will now build and deploy your website. This usually takes 1-3 minutes. You can check the progress by:

1. Going to the **Actions** tab in your repository
2. Looking for a workflow run called "pages build and deployment"
3. When it shows a green checkmark, your site is live!

14. Verify the published result (5 points)

Test the live URL rather than relying only on the local version. Confirm that the final published result matches the merged repository and that links, layout, and content remain correct. If you notice any difference, limitation, or deployment issues, even if the deployment worked correctly, please report them as issues under your GitHub repo.

Part 6. Reflection and AI Accountability

15. Reflection (10 points)

Answer each question by adding another issue in the repo called Reflection, where you put these questions and your answers:

A. GitHub practice. Which GitHub action, issue, branch, commit, pull request, review, or Pages deployment was most useful to you, and why?

B. AI practice. Which AI-generated suggestion did you accept, and what evidence justified accepting it?

C. AI limitation. What did the AI get wrong, overlook, or make unnecessarily complicated? How did you detect and correct it?

D. Engineering judgment. How did the workflow change your understanding of the statement: “AI may generate code, but engineers remain responsible for the system”?

E. Next improvement. Identify one improvement you would make in a second iteration. State its value, risk, and the evidence you would need before implementing it.

Required Submission Package

2 Documents required: Submit a zipped version of your repo (you download it then zip it), and submit one text document containing:

- 1 Repository URL.
 - 2 Live GitHub Pages URL.
 - 3 Original issue URL.
 - 4 Feature-branch URL or commit history URL.
 - 5 Pull-request URL.
 - 6 Merge commit or repository history URL.
 - 7 ai-log.md or an equivalent AI-use record.
 - 8 Acceptance criteria and test results.
 - 9 Screenshots or screen recording for at least three tests.
 - 10 Human code-review table.
 - 11 Reflection responses.
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Grading Rubric

Criterion	Points	Full-credit standard
Acceptance criteria and issue	10	Defines observable goals, audience, non-goal, and task checklist.
Repository and branch setup	15	Creates an appropriate repository, preserves attribution, and uses a feature branch.
AI prompt and planning	15	Supplies context, constraints, acceptance criteria, risks, and a requested plan.
Incremental implementation	10	Produces at least three meaningful, understandable commits.
Human code review	10	Inspects AI output for correctness, privacy, accessibility, security, and maintainability.
Testing and evidence	10	Performs repeatable local and live-site tests and provides evidence.
Pull request and merge workflow	13	Opens a complete PR, conducts review, resolves issues, and merges appropriately.
GitHub Pages deployment	7	Publishes and verifies the live site.
Reflection and accountability	10	Explains accepted output, rejected output, limitations, and engineering judgment.
Total	100	

Changlog

2026_08_02. **Version:** 1.0-Beta. Initiated the assignment